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INTERIM FEDERAL SPECIFICATION

LEAK DETECTOR, REFRIGERANT GAS (KIT)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type of leak detector kit.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Type. The leak detector kit covered by this specification shall be furnished in one type as hereinafter specified.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Set-Up, Paperboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC; Atlanta, Chicago, Kansas City, MO; Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:

H28 - Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

Underwriters' Laboratories, Inc. (UL), Standards:

UL 147 - LP-Gas Torches.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, L.I., NY 11746; 1655 Scott Boulevard, Santa Clara, CA 95050.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, NW., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials. Materials used in the construction of the leak detector kits shall be as specified herein. When materials are not definitely specified, they shall be of good quality normally used for the purpose in commercial practice. All materials shall be new and unused, and free from any defects or imperfections which might affect the appearance or serviceability of the leak detector kits.

3.2 Design and construction. The leak detector shall be so designed and constructed as to be convenient to operate, and entirely adequate for the purpose intended. All parts subject to wear shall be readily accessible for adjustment or repair. Construction shall be free from any defects or characteristics that may render the leak detectors unsuitable or inefficient for the purpose intended.

3.2.1 Interchangeability. All replaceable parts shall be manufactured so that any such part may be replaced without requiring modification.

3.3 UL certification. Each contractor shall submit to the contracting agency, proof that the leak detector he proposes to supply under this specification conforms to the requirements of the Underwriters' Laboratories, Inc., Standard for Safety No. 147, LP - Gas Torches. The label, or listing with reexamination of the Underwriters' Laboratories, Inc., may be accepted as evidence that the leak detector conforms to this requirement. In lieu of the label, or listing with reexamination, the contractor may submit independent proof, satisfactory to the contracting agency, that the leak detector conforms to the applicable requirements of the published standards, including methods of tests of the Underwriters' Laboratories Standards for Safety No. 147, LP - Gas Torches.

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3.4 Compliance with the requirements of 3.1 through 3.3 does not absolve the contractor from complete compliance with the requirements of this specification.

3.5 Leak detector kit. The leak detector kit shall consist essentially of a disposable fuel cylinder (fully loaded), burner assembly, search hose, operating instructions, and carrying case.

3.5.1 Cylinder. The cylinder shall be furnished loaded with a sufficient amount of propane gas to last at least 14 hours when using a flame of proper height for determining refrigerant gas leakage. The cylinder shall be provided with a 1-inch diameter neck with 20 NEF threads per inch for connecting the burner unit.

3.5.2 Burner assembly. The burner unit shall be provided with a reactor plate. The adjustment needle in the regulating valve may be made of steel. A metal tube at least 2 inches long by approximately 5/16 inch outside diameter by 3/16 inch inside diameter shall be attached to the regulating valve. The burner end of the tube shall be externally threaded to receive the burner, consisting of an intake nut, burner head, and chimney. The intake nut shall provide for attachment of the search hose. The chimney shall have an opening for observation of the flame.

3.5.3 Search hose. The search hose shall be of rubber and at least 28 inches in length. Attachment of the hose to the intake nut shall be through a rigid metal tube, or a flexible metal hose, and threaded connection, or similar means which will protect the rubber from the burner heat.

3.5.4 Carrying case. The carrying case shall be of sheet metal not less than .016 inch in thickness before finishing with a good grade of enamel or lacquer. The case shall be of such size as to hold the fuel cylinder, burner assembly, search hose, and any other components or spare parts normally furnished by the contractor in commercial practices. The case shall be equipped with a hinged cover, means for holding the cover in closed position, and a carrying handle.

3.6 Spare parts. In addition to the assembled reactor plate, two reactor plates shall be furnished as spare parts in the case.

3.7 Threads. All threaded parts shall be in the inch system and shall conform to Handbook H28.

3.8 Workmanship. Workmanship of the leak detector kits shall be of the quality prevailing among manufacturers normally producing leak detector kits of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance.

4.2.1 Inspection lot. All leak detector kits offered for delivery at one time shall be considered a lot for the purpose of acceptance, inspection, and tests.

4.2.2 Sampling for visual and dimensional examination. For purposes of examination, as specified in 4.3, samples shall be selected from each lot in accordance with level II of MIL-STD-105. The acceptable quality level (AQL) shall be 1.5 percent defective.

4.2.3 Sampling for tests. One leak detector kit shall be taken from each lot for the test specified in 4.4.

4.3 Visual and dimensional examination. Each of the sample leak detector kits shall be examined for material, dimensions, protective coating, marking, workmanship, and general compliance with this specification. Any leak detector kit in the sample containing one or more defects shall be rejected, and if the number of defective leak detector kits in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

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4.4 Test.

4.4.1 Operational test. The leak detector kits shall operate satisfactorily for locating gas leaks in refrigeration systems. When the search hose is exposed to a gas leak, the escaping gas will be drawn into the tube, hence to the chimney which houses the gas flame. Diffusion of the propane gas flame of the torch by the refrigeration gas from the line leak will cause a change of color of the flame at the reactor plate.

4.5 Rejected lots. Rejected lots may be resubmitted for examination and tests provided the contractor has removed or repaired all nonconforming leak detector kits. Samples shall again be selected and inspected from such resubmitted lots to verify compliance with this specification. Rejected leak detector kits shall not be resubmitted for inspection without furnishing full particulars concerning previous rejections, and measures taken to overcome the defects.

4.6 Inspection of preparation for delivery requirements. An inspection shall be made to determine whether packaging, packing, and marking requirements are in compliance with section 5 of this specification. Defects shall be scored in accordance with Table I. For examination of contents, the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be II of MIL-STD-105 and the AQL shall be 4.0 defects per hundred units.

TABLE I. Classification of preparation for delivery defects

Examine	Defect
Markings (exterior & interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not conforming to the applicable specification or requirements stated therein.
Contents	Quantity per shipping container is more or less than required. Gross weight exceeds the requirements of the box specification.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Each kit shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665, PPP-B-676, or PPP-B-636. The boxes shall be closed in accordance with the applicable box specification or appendix thereto.

5.1.2 Level C. The kits shall be packaged to insure protection during shipment and safe delivery to destination.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The kits packaged as specified in 5.1 shall be packed in quantities as specified (see 6.2) in close-fitting boxes conforming to PPP-B-636, class weather-resistant, or to PPP-B-640, class 2. Closure, waterproof sealing, and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.2 Level B. Twelve kits packaged as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. Closure of the boxes shall be in accordance with method II of the appendix to PPP-B-636.

5.2.3 Level C. The kits packaged as specified in 5.1 shall be packed to comply with the National Motor Freight Classification rules or the Uniform Freight Classification rules.

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5.3 Marking. In addition to any special marking required by the contract or order, and that specified in 5.3.3, all marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. Marking shall be in accordance with MIL-STD-129.

5.3.3 Special marking. In addition to the marking specified above, each shipping container shall be marked with the flash point of the product, expressed in both degrees Fahrenheit and Centigrade; i.e., "Flash Point -- °F. (--°C.)." These markings shall be not less than 1/2 inch high and shall be placed on the same side of the shipping container as the identification markings.

6. NOTES

6.1 Intended use. Leak detectors covered by this specification are intended for use to locate gas leaks in refrigeration systems.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) When special markings are required (see 5.3).

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